

Cheng-Tse Liu (Alex)

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EDUCATION

University of California, Santa Cruz

Santa Clara, CA

- Master of Science in **Natural Language Processing** (Computer Science)
- GPA: 3.9/4.0

Sep. 2023 - Dec. 2024

National University of Kaohsiung

Kaohsiung, Taiwan

- Bachelor of Science in **Information Management**
- GPA: 3.52/4.0, Last 60 credits: 3.8/4.0

Sep. 2018 - Jun. 2022

SKILLS

Technical Skills: Python, Java, C, HTML/CSS, PHP, Flutter, Keras, **Pytorch**, **Tensorflow**, **Scikit-learn**, MySQL, Firebase, Deep Neural Networks, Regression/Classification, **Machine Learning**, NumPy, Pandas, **LLMs**, **BERT**, **Transformer**, **NLP**, **RAG**, **LangChain**, **Microsoft Azure**

PROFESSIONAL EXPERIENCE

Adobe

Santa Clara, CA

Machine Learning Engineer Intern (Advisor: Ryan Rossi, Hanieh Deilamsalehy, Franck Dernoncourt) Mar. 2024 - Dec. 2024

- In the team of **Efficient Summarization of Very Long Documents using Large Language Models**
- Developed a **multi-agent pipeline** for efficiently summarizing long documents using state-of-the-art **LLMs**
- Developed algorithms for text segmentation and pipeline construction
- Deployed and evaluated LLMs including **GPT-4o**, **Llama3**, **Claude**, **Gemini** and **Mistral**

Computational Biology & Intelligence System Lab, National University of Kaohsiung

Kaohsiung, Taiwan

Research Assistant (Advisor: DR. TZU-HSIEN YANG)

Jan. 2021 - Jan. 2022

- Crawled data from websites and preprocessed data as Natural Language Processing (NLP) model input
- **Fine-tuned BERT models** to identify the transcription factor-gene pairs association
- Achieved outstanding accuracy of 83%
- Used the ROC curve analysis technique to evaluate the results of different Internal Ribosome Entry Site (IRES) prediction tools

PROJECT EXPERIENCE

JeBERTy

Group Leader

Jan. 2024 - Mar. 2024

- Fine-tune a pre-trained generative model on answer-questioning Japanese dataset
- Train a BERT model to predicted sentence similarity
- Achieved 0.89 Cosine Similarity for generated questions on test set

Semantic Textual Relatedness for African and Asian Languages

Group Leader

Sep. 2023 - Jan. 2024

- Fine-tuned BERT model on Hausa named entity recognition
- Achieved 0.03 MSE

Recardmend

Major SWE

Mar. 2021 - Jun. 2022

- Crawled data from bank websites and process data for database management
- Designed and visualized the Android APP interface using Flutter and Android Studio
- Built the APP function and user interface for live updated credit card rewards
- The project eventually got the Undergraduate research fellowship, Ministry of Science and Technology, Taiwan

WebDBLink: Integrated Database Management and Web Interface System

Major SWE

May. 2021 - Jan. 2022

- Leveraged ERD, Relational Model, UML and Normalization to systematically construct databases
- Filtered data with SQL queries and presented the searching result on a website

PUBLICATIONS

- T.-H. Yang*, C.-Y. Wang†, H.-C. Tsai†, Y.-C. Yang† and C.-T. Liu, “YTLR: a web service for automatic yeast transcription factor-gene association extraction from the literature” Computational and Structural Biotechnology Journal, vol. 20, pp. 4636-4644, 2022 (SCI 2022 impact factor = 6.155, Ranking 23.6% (70/296) in Biochemistry Molecular Biology)
- T.-H. Yang*, C.-Y. Wang†, H.-C. Tsai†, and C.-T. Liu†, “Human IRES Atlas: an integrative platform for studying IRES-driven translational regulation in humans” Database, vol. 2021: article ID baab025; doi:10.1093/database/baab025, 2021. (SCI 2020 impact factor = 3.451, Ranking 24% (14/58) in Mathematical Computational Biology)